IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

OPTIMAL CONTROL

Thursday, 15 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

This is an OPEN-BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S.A. Evangelou
Second Marker(s) : G. Scarciotti

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OPTIMAL CONTROL

1. Consider the scalar linear system with one control variable

$$\dot{x}(t) = u(t)$$

which is to be controlled so that the following cost functional is minimised

$$J(u) = \int_0^2 \left(x(t) - t^2\right)^2 \, dt$$

with $x(0) = 1$ and $x(2)$ free, and $u(t) \in [0,4]$ for all t.

a) Construct the Hamiltonian and write the switching function of the problem. [3]

b) Derive the adjoint equation and boundary conditions of the problem. [3]

c) Investigate the existence of singular arcs and calculate the optimal control law in terms of the switching function. [6]

d) It can be shown by contradiction that the optimal solution satisfies $x^*(t) \geq t^2$ for $t \in [0,2]$. Compute as a function of time

i) the optimal control, [4]

ii) the optimal state trajectory. [4]

2. The motion of a boat moving at constant forward speed in a horizontal xy-plane can be described by a simple model as follows

$$\dot{x}(t) = V\cos\psi(t)$$
$$\dot{y}(t) = V\sin\psi(t)$$

where $x(t)$ and $y(t)$ are the Cartesian coordinates of the boat position in the plane, V is the constant forward speed and $\psi(t)$ is the heading angle, as shown in Figure 2.1.

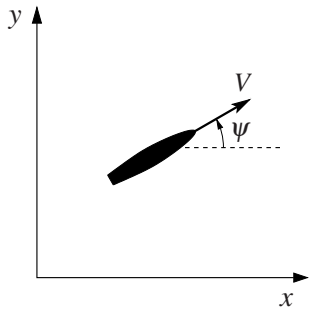


Figure 2.1 Boat travelling at constant forward speed.

It is desired to make a fishing trip from the initial position

$$x(0) = 0, \qquad y(0) = 0$$

to the final position

$$x(2) = 1, \qquad y(2) = 0$$

by manipulating the heading angle $\psi(t)$. It is known that there is more fish at positions with higher y-coordinates, therefore, the trip is organised to maximise

$$\int_0^2 y(t)\,dt$$

The admissible boat position and heading angle values are not constrained.

a) Write the state and control variables of the problem. [2]

b) Write the functional $J(u)$ that should be minimised, where u is the control variable. [2]

c) Construct the Hamiltonian of the problem. [2]

d) By considering the maximisation of the Hamiltonian, derive the optimal control in terms of the states and costates (Hint: write the optimal control in terms of $\tan u^*(t)$). [3]

e) Derive the adjoint equations. [2]

f) Calculate the optimal control $\tan u^*(t)$ as a function of time and in terms of constant parameters. [2]

g) Compute the optimal state trajectory as a function of time t and in terms of constant parameters, and use that to calculate the general form of the extremal $y(x)$ curve followed by the boat on the xy-plane. Hint: $\sin(\arctan z) = \frac{z}{\sqrt{1+z^2}}$, $\cos(\arctan z) = \frac{1}{\sqrt{1+z^2}}$, $\int \frac{1}{\sqrt{1+z^2}}\,dz = \ln\left|z+\sqrt{z^2+1}\right| + C$. [7]

3. As in Q2, the motion of a boat moving at constant forward speed in a horizontal xy-plane can be described by a simple model as follows

$$\dot{x}(t) = V\cos\psi(t)$$
$$\dot{y}(t) = V\sin\psi(t)$$

where $x(t)$ and $y(t)$ are the Cartesian coordinates of the boat position in the plane, V is the constant forward speed and $\psi(t)$ is the heading angle, as shown in Figure 3.1.

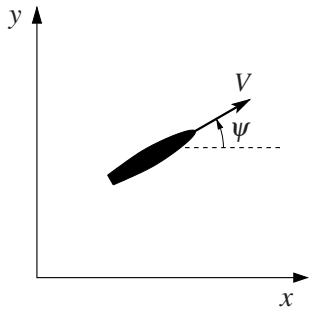


Figure 3.1 Boat travelling at constant forward speed.

It is desired to make a fishing trip from the initial position

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$$x(2) = 1, \qquad y(2) = 0$$

by manipulating the heading angle $\psi(t)$. It is known that there is more fish at positions with higher y-coordinates, therefore, the trip is organised to maximise

$$\int_0^2 y(t)\,dt$$

The admissible boat position and heading angle values are not constrained.

Formulate and solve the problem as a calculus of variations problem by answering the steps below. (Hint: first write the problem as an optimal control problem with respect to time t and then substitute the time to write the problem with respect to the position coordinate x; also, recall that $V = \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2}$).

a) Write the functional $J(y)$ that should be minimised. [5]

b) Calculate the boundary conditions of the problem. [2]

c) Write the constraint of the problem. [4]

d) Write the first-order necessary condition for optimality given by the Euler-Lagrange equation. [6]

e) Compute the general form of the extremal $y(x)$ curve. [3]

4. Consider the Linear Quadratic Regulator (LQR) tracking problem in which:

The control system is a linear time-varying system

$$\dot{x} = A(t)x + B(t)u, \qquad x(t_0) = x_0$$

with $x \in \mathbb{R}^n$ and $u \in \mathbb{R}^m$ (the control is unconstrained).

The target set is $S = \{t_f\} \times \mathbb{R}^n$, where t_f is a fixed time (fixed-time, free-endpoint problem).

The cost functional to be minimised is

$$J(u) = \int_{t_0}^{t_f} \Big((x(t) - x_{ref}(t))^T Q(t) (x(t) - x_{ref}(t)) + u^T(t) R(t) u(t) \Big) dt + (x(t_f) - x_{ref}(t_f))^T M (x(t_f) - x_{ref}(t_f))$$

where $x_{ref}(t) \in \mathbb{R}^n$ is a given reference trajectory and $Q(t)$, $R(t)$, M are matrices of appropriate dimensions satisfying $M = M^T \geq 0$, $Q(t) = Q^T(t) \geq 0$, and $R(t) = R^T(t) > 0$ for all $t \in [t_0, t_f]$.

a) Write the associated Hamilton-Jacobi-Bellman (HJB) differential equation. [3]

b) Compute the control in terms of the value function that achieves the infimum in the HJB equation and use the computed control to simplify the HJB equation. [3]

c) Write the boundary condition that must be satisfied by the value function. [3]

d) Assume that the HJB differential equation admits a solution of the form

$$V(t,x) = x^T P(t) x + 2x^T b(t) + c(t)$$

where $P(t)$, $b(t)$ and $c(t)$ are unknown functions of t of appropriate dimensions.

i) Calculate $P(t_f)$, $b(t_f)$, and $c(t_f)$ so that the assumed solution matches the boundary condition found in part c). [3]

ii) Derive a system of ordinary differential equations for $P(t)$, $b(t)$, and $c(t)$, such that the value function $V(t,x)$ satisfies the HJB equation. [6]

iii) Compute the optimal control law. [2]